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Mrittika Roy

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Work Experiences

Ztrios Technologies Ltd., AI Developer — September 2025 to Current

- Implemented **RAG(Retrieval Augmented Generation** in different AI projects that support Bengali
- Performed data preprocessing and cleaning for noisy data
- Fine-tuned Qwen3-4B using PEFT(LoRA) of rank 16
- Conducted R&D on different AI-driven applications

Samsung R&D Institute, Bangladesh, Software Test Intern — July 2024 to July 2025

- Gained hands-on experience in software testing methodologies and tools
- Developed and executed automated UI test scripts for the Samsung Notes application using **Selenium**
- Worked on preparing a DOI for **patent** based on Image Processing

Education

BSc Rajshahi University of Engineering & Technology, Rajshahi, CSE — Feb. 2019 to May 2024

- CGPA: 3.66/4.0
- **Courseworks:** C Programming, Discrete Mathematics, Data Structure, OOP, Algorithms, Database Management, Applied Statistics & Queuing Theory, Computer Architecture, Computer Network, Microprocessor and Assembly Language, Artificial Intelligence, Digital Signal Processing, Digital Image Processing, Compiler Design, Neural Networks and Fuzzy Systems, Network Security, Data Mining, VLSI Design.

HSC Holy Cross College, Dhaka — June 2016 to July 2018

- GPA: 5.0/5.0

SSC Jamalpur Govt. Girls' High School, Jamalpur — Jan. 2006 to May 2016

- GPA: 5.0/5.0

Research Projects

BGCA-Fusion: Bidirectional Gated Cross-Attention for Multimodal Breast Cancer Diagnosis from Mammograms and Clinical Reports (Under-review on "Advanced Intelligent Systems" [IF 6.1] of Wiley)

- Three different mammography datasets were used in this experiment.
- Image and textual features extracted using the CNN, ViT and LLM are fused for the classification task.
- Employed XAI techniques for evaluating model interpretability.

Public Sentiment Analysis of the Israel-Palestine Conflict: A Reddit-Based Study

- 'Daily Public Opinion on Israel-Palestine War' dataset was labeled using VADER.
- A hybrid model consisting of RoBERTa and BI-LSTM model is trained from scratch on the dataset for sentiment analysis. Also DistilBERT model is fine tuned on this dataset for experiment purpose. For ablation study experiment is done on Logistic Regression model.
- Link: `https://ieeexplore.ieee.org/abstract/document/11172186`

Text to Image Generation Using GAN Based on XLNet (Undergraduation Thesis)

- Text Analysis: Leveraged XLNet's permutation-based modeling to extract high-dimensional linguistic features, bridging autoencoder and autoregressive NLP techniques.
- Image Synthesis: Implemented DF-GAN to generate high-fidelity imagery through the deep fusion of textual and visual feature maps.

Explainable-Dual-Stage-RAG(On-going)

- Developed a medical Q&A system using Mistral-7B/LLaMA-3 that separates rationale generation from evidence-based synthesis to improve reasoning depth.
- Integrated Contriever-based dense search and BM25, fused via Reciprocal Rank Fusion (RRF) and Cross-Encoder reranking, for high-precision medical document retrieval.

AI-generated Image Detection Using Contrastive Learning(On-going)

- Made a dataset of real-fake images. Real mages are taken from the MSCOCO dataset, and the fake images are generated using the captions in the MSCOCO dataset
- Conducting training for the contrastive learning approach with frequency domain features with supcon loss
- Applying cross-dataset validation to advance the generalization to the production level

Projects

Dikkha AI

- A curriculum-aligned Bengali educational Q&A app supporting 20+ NCTB subjects, featuring a subject-aware query-rewriting agent and RAG-based retrieval pipeline.
- **Tech Stack:** Python, FastAPI, Supabase, Google Gemini, GCP Cloud Run
- **Play Store Link::** https://play.google.com/store/apps/details?id=com.dikkhaai.app&pli=1

Chhar - Scrapper Agent

- An automated web scraping agent that discovers retail discount offers across Dhaka, extracts and structures them using LLMs, and publishes enriched data to a production backend API.
- **Tech Stack:** Tavily, Firecrawl, Google Gemini, Google Maps Platform, Google Cloud Storage, FastAPI
- **Play Store Link:** https://play.google.com/store/apps/details?id=dev.neuronfish.chhar&hl=en

Visual Search

- A CPU-friendly catalog image search system using DINOv2 ViT-B/14 for feature extraction, FAISS for similarity retrieval, and optional text-guided filtering and zero-shot object cropping via OWL-ViT.
- **Tech Stack:** DINOv2, FAISS, OWL-ViT, FastAPI, spaCy, timm
- **Github Link:** https://github.com/Mrittika6026/visual_search

Technologies

Languages: Python, C, C++, HTML5/CSS, SQL

Applications: Visual Studio Code, Codeblocks, Github, Google Colab, Kaggle, Jupyter Notebook, Cisco Packet Tracer, Logism, Digital, Overleaf

Frameworks: FastAPI, PyTorch, LangGraph

Awards

Dhaka Board HSC scholarship: Received general grade scolarship with fullfree tuition fee and stipend in Bachelor.

Dhaka Board SSC scholarship: Received talent full scolarchip with full free tuition fee and stipend in Higher Secondary.

Perfect Attendance Certificate Received for 100% attendance throughout higher secondary studies.